



12571 - N6946-BH1: Fading following a failed supernova, or post-merger recovery?

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
MIRI				
	2	MIRI	MIRI Imaging	(1) NAME-N6946-BH1
	3	NIRCam	NIRCam Imaging	(1) NAME-N6946-BH1

ABSTRACT

Recent JWST imaging revealed a luminous infrared source at the location of failed supernova (SN) candidate N6946-BH1. While significantly fainter than the progenitor (13-25% of the bolometric luminosity), the spectral energy distribution (SED) is consistent with both a failed SN and a stellar merger or outburst event. Curiously, observations taken one month later in matching filters revealed a 50% flux reduction in the shortest MIRI filter (5.6 μ m). The mid-IR fluxes also appear to resemble Case C Polycyclic Aromatic Hydrocarbon (PAH) emission, typically produced when dust is irradiated by a UV field.

We are requesting a total of 16 hours of JWST time in Cycle 4: 5 hours for continued monitoring of N6946-BH1 to assess variability across the near- and mid-IR, and 11 hours for medium-resolution MIRI spectroscopy to search for PAH features. In addition, we are requesting another 5 hours in

Cycle 5 to repeat the photometry observations so that we can determine the rate of change in brightness of the source. If the near-IR flux continues to fade, it will strongly suggest the formation of a black hole. If the source remains constant or brightens in the near-IR while fading in the mid-IR, this will support a merger scenario, with the dust clearing to reveal the surviving star.

These observations will not only constrain the timescales for such events but also help estimate the ejected dust mass from mid-IR emission, ultimately resolving a long-standing debate over the fate of N6946-BH1.

OBSERVING DESCRIPTION

We are proposing to image the remaining infrared source remaining at the position of failed SN candidate N6946-BH1 with NIRCам and MIRI over two epochs. This will enable us to understand how the source changes in brightness across the SED with time, and will allow us to constrain the total dust mass of the system. Ultimately these observations will constrain the fate of BH1: has it formed a black hole, or is there a surviving star following a merger?

We are also requesting 10 hours of JWST time to obtain a MIRI MRS spectrum of the remaining source at the position of failed SN candidate BH1. The MRS spectrum will reveal whether or not there are PAH features present in the SED, and break the degeneracy between the merger and failed SN scenarios.

Proposal 12571 - Targets - N6946-BH1: Fading following a failed supernova, or post-merger recovery?

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d) Dec: +60 08 8.29 (60.13564d) Equinox: J2000	Epoch of Position: 2000	
<i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Star</i> <i>Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]</i>					

Proposal 12571 - Observation 2 - N6946-BH1: Fading following a failed supernova, or post-merger recovery?

Observation	Proposal 12571, Observation 2: MIRI										Thu May 14 23:00:08 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: MIRI Imaging										
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d) Dec: +60 08 8.29 (60.13564d) Equinox: J2000			Epoch of Position: 2000					
Template	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Star Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]										
Dithers	Subarray										
	FULL										
Spectral Elements	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	4		1	1			DEFAULT	
Special Requirements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	21	1	1	Dither 1	4	4	233.103	
	2	F770W	FASTR1	11	1	1	Dither 1	4	4	122.102	
	3	F1000W	FASTR1	11	1	1	Dither 1	4	4	122.102	
	4	F1280W	FASTR1	11	1	1	Dither 1	4	4	122.102	
	5	F1800W	FASTR1	11	1	1	Dither 1	4	4	122.102	
	6	F2100W	FASTR1	11	1	1	Dither 1	4	4	122.102	
	7	F2550W	FASTR1	11	1	1	Dither 1	4	4	122.102	
Special Requirements	On Hold Observation ON Hold util completion of 7648. Once complete, 12571 should be scheduled ~1 year later										

Proposal 12571 - Observation 3 - N6946-BH1: Fading following a failed supernova, or post-merger recovery?

Observation	Proposal 12571, Observation 3: NIRCcam										Thu May 14 23:00:08 GMT 2026
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCcam Imaging										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	NAME-N6946-BH1	RA: 20 35 27.5600 (308.8648333d)			Epoch of Position: 2000					
			Dec: +60 08 8.29 (60.13564d)								
			Equinox: J2000								
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.								
Template	Category=Star										
	Description=[Black holes, Circumstellar dust, M supergiants, Red supergiants]										
Module	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRASCA		4		STANDARD		8" (SMALL)		1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F250M	MEDIUM8	10	1	4	4	4208.814		
	2	F182M	F360M	SHALLOW4	10	1	4	4	2104.407		
Special Requirements	On Hold Observation On Hold until completion of 7648. Once complete, 12571 should be scheduled ~1 year later										